

HW 10

7.4

Suppose that X and Y are independent exponential random variables with parameters $\lambda \neq \mu$. Find the density function of $X + Y$.

✓ Answer ✓

$$\int_0^s \lambda e^{-\lambda x} \mu e^{-\mu(s-x)} dx = \lambda \mu e^{-\mu s} \frac{1 - e^{-(\lambda-\mu)s}}{\lambda - \mu} = \frac{\lambda \mu}{\lambda - \mu} (e^{-\mu s} - e^{-\lambda s})$$

7.6

We deal 5 cards from a standard deck of 52, one by one. What is the probability that the third is a king, and the fifth is the ace of spades?

✓ Answer

$$\frac{4}{52} \cdot \frac{1}{51} = \frac{1}{663}$$

7.16

Let X be a Poisson random variable with parameter λ and Y an independent Bernoulli random variable with parameter p . Find the probability mass function of $X + Y$.

✓ Answer

$$S = X + Y$$

$$\text{For } S = 0, P(S = s) = (1 - p)e^{-\lambda}$$

$$\text{Else, } (1 - p) \frac{e^{-\lambda} \lambda^s}{s!} + p \frac{e^{-\lambda} \lambda^{(s-1)}}{(s-1)!} = e^{-\lambda} \lambda^s s! \left(1 - p + \frac{sp}{\lambda}\right)$$

7.22

Let X and Y be two independent normals with the same $N(\mu, \sigma^2)$ distribution, with μ real and $\sigma^2 > 0$. Are there values μ and σ^2 for which $2X$ and $X + Y$ have the same distribution?

✓ Answer

$X + Y$ will always have more variance than $2X$ since we sample twice, instead of doubling our sample repeatably, however if our initial distribution has no variance, then our outputs should be equal.

If we ignore the $\sigma^2 > 0$ restriction, then μ is anything, $\sigma^2 = 0$
Otherwise, no

7.28

Let X_1, X_2, X_3 be independent $\text{Exp}(\lambda)$ distributed random variables. Find the probability that $P(X_1 < X_2 < X_3)$

✓ Answer

Since all 3 are symmetrical continuous variables, we know all orderings of 3 samples are equally likely, making this ordering have a chance of $\frac{1}{6}$

8.2

Caleb has a four-sided die, a six-sided die, and a 12-sided die. He rolls the three dice and adds up the numbers showing. What is the expected value of the sum of the rolls?

✓ Answer

$$A \sim \text{Uniform}(1, 4)$$

$$B \sim \text{Uniform}(1, 6)$$

$$C \sim \text{Uniform}(1, 12)$$

$$S = A + B + C$$

$$\begin{aligned} E[S] &= E[A + B + C] = E[A] + E[B] + E[C] \\ &= 2.5 + 3.5 + 6.5 = 12.5 \end{aligned}$$

8.8

Our faucet is broken, and a plumber has been called. The arrival time of the plumber is uniformly distributed between 1PM and 7PM. Independently of when the plumber arrives, the time it takes to fix the broken faucet is exponentially distributed with mean 30 minutes. Find the expectation and variance of the time at which the plumber completes the project.

✓ Answer

$$A \sim \text{Uniform}(1, 7)$$

$$T \sim \text{Exp}(\beta = 0.5)$$

$$C = A + T$$

$$\begin{aligned} E[C] &= E[A + T] = E[A] + E[T] \\ &= 4 + 0.5 = 4:30 \text{ PM} \end{aligned}$$